9 Classical Conditioning and Reinforcement Learning

9.1 Introduction

The ability of animals to learn appropriate actions in response to particular stimuli on the basis of associated rewards or punishments is a focus of behavioral psychology. The field is traditionally separated into classical (or Pavlovian) and instrumental (or operant) conditioning. In classical conditioning, the reinforcers (i.e., the rewards or punishments) are delivered independently of any actions taken by the animal. In instrumental conditioning, the actions of the animal determine what reinforcement is provided. Learning about stimuli or actions solely on the basis of the rewards and punishments associated with them is called reinforcement learning. Reinforcement learning is minimally supervised because animals are not told explicitly what actions to take in particular situations, but must work this out for themselves on the basis of the reinforcement they receive.

classical and instrumental conditioning

reinforcement learning

We begin this chapter with a discussion of aspects of classical conditioning and the models that have been developed to account for them. We first discuss various pairings of one or more stimuli with presentation or denial of a reward, and present a simple learning algorithm that summarizes the results. We then present an algorithm, called temporal difference learning, that leads to predictions of both the presence and the timing of rewards delivered after a delay following stimulus presentation. Two neural systems, the cerebellum and the midbrain dopamine system, have been particularly well studied from the perspective of conditioning. The cerebellum has been studied in association with eyeblink conditioning, a paradigm in which animals learn to shut their eyes just in advance of disturbances, such as puffs of air, that are signaled by cues. The midbrain dopaminergic system has been studied in association with reward learning. We focus on the latter, together with a small fraction of the extensive behavioral data on conditioning.

There are two broad classes of instrumental conditioning tasks. In the first class, which we illustrate with an example of foraging by bees, the reinforcement is delivered immediately after the action is taken. This makes learning relatively easy. In the second class, the reward or punishment depends on an entire sequence of actions and is partly or wholly delayed

delayed rewards

until the sequence is completed. Thus, learning the appropriate action at each step in the sequence must be based on future expectation, rather than immediate receipt, of reward. This makes learning more difficult. Despite the differences between classical and instrumental conditioning, we show how to use the temporal difference model we discuss for classical conditioning as the heart of a model of instrumental conditioning when rewards are delayed.

For consistency with the literature on reinforcement learning, throughout this chapter, the letter r is used to represent a reward rather than a firing rate. Also, for convenience, we consider discrete actions such as a choice between two alternatives, rather than a continuous range of actions. We also consider trials that consist of a number of discrete events and use an integer time variable $t = 0, 1, 2, \ldots$ to indicate steps during a trial. We therefore also use discrete weight update rules (like those we discussed for supervised learning in chapter 8) rather than learning rules described by differential equations.

9.2 Classical Conditioning

Classical conditioning involves a wide range of different training and testing procedures and a rich set of behavioral phenomena. The basic procedures and results we discuss are summarized in table 9.1. Rather than going through the entries in the table at this point, we introduce a learning algorithm that serves to summarize and structure these results.

unconditioned stimulus and response

conditioned stimulus and response

In the classic Pavlovian experiment, dogs are repeatedly fed just after a bell is rung. Subsequently, the dogs salivate whenever the bell sounds, as if they expect food to arrive. The food is called the unconditioned stimulus. Dogs naturally salivate when they receive food, and salivation is thus called the unconditioned response. The bell is called the conditioned stimulus because it elicits salivation only under the condition that there has been prior learning. The learned salivary response to the bell is called the conditioned response. We do not use this terminology in the following discussion. Instead, we treat those aspects of the conditioned responses that mark the animal's expectation of the delivery of reward, and build models of how these expectations are learned. We therefore refer to stimuli, rewards, and expectation of reward.

Predicting Reward: The Rescorla-Wagner Rule

The Rescorla-Wagner rule (Rescorla and Wagner, 1972), which is a version of the delta rule of chapter 8, provides a concise account of certain aspects of classical conditioning. The rule is based on a simple linear prediction of the reward associated with a stimulus. We use a binary variable u to represent the presence or absence of the stimulus ($u = 1$ if the stimulus

Paradigm	Pre-Train	Train	Result
Pavlovian		$s \to r$	$s \to$ 'r'
Extinction	$s \to r$	$s \to \cdot$	$s \to$ '$\cdot$'
Partial		$s \to r \quad s \to \cdot$	$s \to \alpha$ 'r'
Blocking	$s_1 \to r$	$s_1 + s_2 \to r$	$s_1 \to$ 'r' $\quad s_2 \to$ '$\cdot$'
Inhibitory		$s_1 + s_2 \to \cdot \quad s_1 \to r$	$s_1 \to$ 'r' $\quad s_2 \to -$'r'
Overshadow		$s_1 + s_2 \to r$	$s_1 \to \alpha_1$'r' $\quad s_2 \to \alpha_2$'r'
Secondary	$s_1 \to r$	$s_2 \to s_1$	$s_2 \to$ 'r'

Table 9.1 Classical conditioning paradigms. The columns indicate the training procedures and results, with some paradigms requiring a pre-training as well as a training period. Both training and pre-training periods consist of a moderate number of training trials. The arrows represent an association between one or two stimuli (s, or s_1 and s_2) and either a reward (r) or the absence of a reward ($\cdot$). In Partial and Inhibitory conditioning, the two types of training trials that are indicated are alternated. In the Result column, the arrows represent an association between a stimulus and the expectation of a reward ('r') or no reward ('$\cdot$'). The factors of α denote a partial or weakened expectation, and the minus sign indicates the suppression of an expectation of reward.

is present, $u = 0$ if it is absent). The expected reward, denoted by v, is expressed as this stimulus variable multiplied by a weight w,

stimulus u
expected reward v
weight w

$$v = wu \ . \tag{9.1}$$

The value of the weight, w, is established by a learning rule designed to minimize the expected squared error between the actual reward r and the prediction v, $\langle (r - v)^2 \rangle$. The angle brackets indicate an average over the presentations of the stimulus and reward, either or both of which may be stochastic. As we saw in chapter 8, stochastic gradient descent in the form of the delta rule is one way of minimizing this error. This results in the trial-by-trial learning rule known as the Rescorla-Wagner rule,

Rescorla-Wagner rule

$$w \to w + \epsilon \delta u \quad \text{with} \quad \delta = r - v \ . \tag{9.2}$$

Here ϵ is the learning rate, which can be interpreted in psychological terms as the associability of the stimulus with the reward. The crucial term in this learning rule is the prediction error, δ. In a later section, we interpret the activity of dopaminergic cells in the ventral tegmental area (VTA) as encoding a form of this prediction error. If ϵ is sufficiently small and $u = 1$ on every trial (the stimulus is always presented), the rule ultimately makes w fluctuate about the equilibrium value $w = \langle r \rangle$, at which point the average value of δ is 0.

associability

The filled circles in figure 9.1 show how learning progresses according to the Rescorla-Wagner rule during the acquisition and extinction phases of Pavlovian conditioning. In this example, the stimulus and reward were both initially presented on each trial, but later the reward was removed. The weight approaches the asymptotic limit $w = r$ exponentially during the rewarded phase of training (conditioning), and exponentially decays to $w = 0$ during the unrewarded phase (extinction). Experimental learning

acquisition
extinction

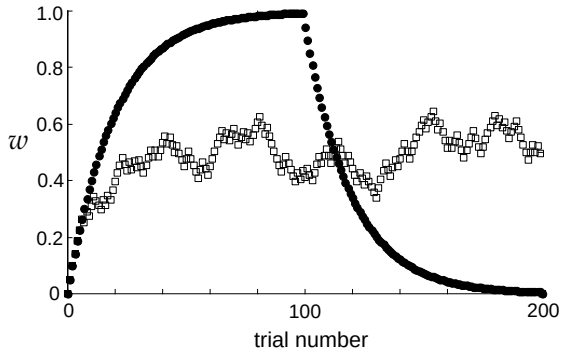

Figure 9.1 Acquisition and extinction curves for Pavlovian conditioning and partial reinforcement as predicted by the Rescorla-Wagner model. The filled circles show the time evolution of the weight w over 200 trials. In the first 100 trials, a reward of $r = 1$ was paired with the stimulus, while in trials 100-200 no reward was paired ($r = 0$). Open squares show the evolution of the weights when a reward of $r = 1$ was paired with the stimulus randomly on 50% of the trials. In both cases, $\epsilon = 0.05$.

curves are generally more sigmoidal in shape. There are various ways to account for this discrepancy, the simplest of which is to assume a nonlinear relationship between the expectation v and the behavior of the animal.

partial reinforcement

The Rescorla-Wagner rule also accounts for aspects of the phenomenon of partial reinforcement, in which a reward is associated with a stimulus only on a random fraction of trials (table 9.1). Behavioral measures of the ultimate association of the reward with the stimulus in these cases indicate that it is weaker than when the reward is always presented. This is expected from the delta rule, because the ultimate steady-state average value of $w = \langle r \rangle$ is smaller than r in this case. The open squares in figure 9.1 show what happens to the weight when the reward is associated with the stimulus 50% of the time. After an initial rise from 0, the weight varies randomly around an average value of 0.5.

stimulus vector $\mathbf{u}$
weight vector $\mathbf{w}$

To account for experiments in which more than one stimulus is used in association with a reward, the Rescorla-Wagner rule must be extended to include multiple stimuli. This is done by introducing a vector of binary variables $\mathbf{u}$, with each of its components representing the presence or absence of a given stimulus, together with a vector of weights $\mathbf{w}$. The expected reward is then the sum of each stimulus parameter multiplied by its corresponding weight, written compactly as a dot product,

$$v = \mathbf{w} \cdot \mathbf{u} \ . \tag{9.3}$$

delta rule

Minimizing the prediction error by stochastic gradient descent in this case gives the delta learning rule,

$$\mathbf{w} \to \mathbf{w} + \epsilon \delta \mathbf{u} \quad \text{with} \quad \delta = r - v \, . \tag{9.4}$$

blocking

Various classical conditioning experiments probe the way that predictions are shared between multiple stimuli (see table 9.1). Blocking is the

paradigm that first led to the suggestion of the delta rule in connection with classical conditioning. In blocking, two stimuli are presented together with the reward, but only after an association has already developed for one stimulus by itself. In other words, during the pre-training period, a stimulus is associated with a reward, as in Pavlovian conditioning. Then, during the training period, a second stimulus is present along with the first, in association with the same reward. In this case, the pre-existing association of the first stimulus with the reward blocks an association from forming between the second stimulus and the reward. Thus, after training, a conditioned response is evoked only by the first stimulus, not by the second. This follows from the vector form of the delta rule, because training with the first stimulus makes $w_1 = r$. When the second stimulus is presented along with the first, its weight starts out at $w_2 = 0$, but the prediction of reward $v = w_1 u_1 + w_2 u_2$ is still equal to r. This makes $\delta = 0$, so no further weight modification occurs.

inhibitory conditioning

A standard way to induce inhibitory conditioning is to use trials in which one stimulus is shown in conjunction with the reward in alternation with trials in which that stimulus and an additional stimulus are presented in the absence of reward. In this case, the second stimulus becomes a conditioned inhibitor, predicting the absence of reward. This can be demonstrated by presenting a third stimulus that also predicts reward, in conjunction with the inhibitory stimulus, and showing that the net prediction of reward is reduced. It can also be demonstrated by showing that subsequent learning of a positive association between the inhibitory stimulus and reward is slowed. Inhibition emerges naturally from the delta rule. Trials in which the first stimulus is associated with a reward result in a positive value of w_1. Over trials in which both stimuli are presented together, the net prediction $v = w_1 + w_2$ comes to be 0, so w_2 is forced to be negative.

overshadowing

A further example of the interaction between stimuli is overshadowing. If two stimuli are presented together during training, the prediction of reward is shared between them. After application of the delta rule, $v = w_1 + w_2 = r$. However, the prediction is often shared unequally, as if one stimulus is more salient than the other. Overshadowing can be encompassed by generalizing the delta rule so that the two stimuli have different learning rates (different values of ϵ), reflecting unequal associabilities. Weight modification stops when $\langle \delta \rangle = 0$, at which point the faster growing weight will be larger than the slower growing weight. Various, more subtle effects come from having not only different but also modifiable learning rates, but these lie beyond the scope of our account.

The Rescorla-Wagner rule, binary stimulus parameters, and linear reward prediction are obviously gross simplifications of animal learning behavior. Yet they summarize and unify an impressive amount of classical conditioning data and are useful, provided their shortcomings are fully appreciated. As a reminder of this, we point out one experiment, secondary conditioning, that cannot be encompassed within this scheme.

secondary conditioning

Secondary conditioning involves the association of one stimulus with a reward, followed by an association of a second stimulus with the first stimulus (table 9.1). This causes the second stimulus to evoke expectation of a reward with which it has never been paired (although if pairings of the two stimuli without the reward are repeated too many times, the result is extinction of the association of both stimuli with the reward). The delta rule cannot account for the positive expectation associated with the second stimulus. Indeed, because the reward does not appear when the second stimulus is presented, the delta rule would cause w_2 to become negative. In other words, in this case, the delta rule would predict inhibitory, not excitatory, secondary conditioning. Secondary conditioning is related to the problem of delayed rewards in instrumental conditioning that we discuss later in this chapter.

Secondary conditioning raises the important issue of keeping track of the time within a trial in which stimuli and rewards are present. This is evident because a positive association with the second stimulus is reliably established only if it precedes the first stimulus in the trials in which they are paired. If the two stimuli are presented simultaneously, the result may be inhibitory rather than secondary conditioning.

Predicting Future Reward: Temporal Difference Learning

We measure time within a trial using a discrete time variable t, which falls in the range $0 \leq t \leq T$. The stimulus $u(t)$, the prediction $v(t)$, and the reward $r(t)$ are all expressed as functions of t.

In addition to associating stimuli with rewards and punishments, animals can learn to predict the future time within a trial at which reinforcement will be delivered. We might therefore be tempted to interpret $v(t)$ as the reward predicted to be delivered at time step t. However, Sutton and Barto (1990) suggested an alternative interpretation of $v(t)$ that provides a better match to psychological and neurobiological data, and suggests how animals might use their predictions to optimize behavior when rewards are delayed. The suggestion is that the variable $v(t)$ should be interpreted as a prediction of the total future reward expected from time t onward to the end of the trial, namely

total future reward

$$\left\langle \sum_{\tau=0}^{T-t} r(t+\tau) \right\rangle . \tag{9.5}$$

The brackets denote an average over trials. This quantity is useful for optimization, because it summarizes the total expected worth of the current state. To approximate $v(t)$, we generalize the linear relationship used for classical conditioning, equation 9.3. For the case of a single time-dependent stimulus $u(t)$, we write

$$v(t) = \sum_{\tau=0}^{t} w(\tau) u(t-\tau) . \tag{9.6}$$

This is just a discrete time version of the sort of linear filter used in chapters 1 and 2.

Arranging for $v(t)$ to predict the total future reward would appear to require a simple modification of the delta rule we have discussed previously,

$$w(\tau) \to w(\tau) + \epsilon\delta(t)u(t-\tau)\,, \tag{9.7}$$

with $\delta(t)$ being the difference between the actual and predicted total future reward, $\delta(t) = \sum_\tau r(t+\tau) - v(t)$. However, there is a problem with applying this rule in a stochastic gradient descent algorithm. Computation of $\delta(t)$ requires knowledge of the total future reward on a given trial. Although $r(t)$ is known at time t, the succeeding $r(t+1), r(t+2)\ldots$ have yet to be experienced, making it impossible to calculate $\delta(t)$. A possible solution is suggested by the recursive formula

$$\sum_{\tau=0}^{T-t} r(t+\tau) = r(t) + \sum_{\tau=0}^{T-t-1} r(t+1+\tau)\,. \tag{9.8}$$

The temporal difference model of prediction is based on the observation that $v(t+1)$ provides an approximation of the average value (across trials) of the last term in equation 9.8, so we can write

$$\sum_{\tau=0}^{T-t} r(t+\tau) \approx r(t) + v(t+1)\,. \tag{9.9}$$

Replacing the sum in the equation $\delta(t) = \sum_\tau r(t+\tau) - v(t)$ by this approximation gives the temporal difference learning rule,

temporal difference rule

$$w(\tau) \to w(\tau) + \epsilon\delta(t)u(t-\tau) \quad \text{with} \quad \delta(t) = r(t) + v(t+1) - v(t)\,. \tag{9.10}$$

The name of the rule comes from the term $v(t+1) - v(t)$, which is the difference between two successive estimates. $\delta(t)$ is usually called the temporal difference error. Under a variety of circumstances, this rule is likely to converge to make the correct predictions.

Figure 9.2 shows what happens when the temporal difference rule is applied during a training period in which a stimulus appears at time $t = 100$, and a reward is given for a short interval around $t = 200$. Initially, $w(\tau) = 0$ for all τ. Figure 9.2A shows that the temporal difference error starts off being nonzero only at the time of the reward, $t = 200$, and then, over trials, moves backward in time, eventually stabilizing around the time of the stimulus, where it takes the value 2. This is equal to the (integrated) total reward provided over the course of each trial. Figure 9.2B shows the behavior during a trial of a number of variables before and after learning. After learning, the prediction $v(t)$ is 2 from the time the stimulus is first presented ($t = 100$) until the time the reward starts to be delivered. Thus, the temporal difference prediction error (δ) has a spike at $t = 99$. This spike persists, because $u(t) = 0$ for $t < 100$. The temporal difference term (Δv) is negative around $t = 200$, exactly compensating for the delivery of reward, and thus making $\delta = 0$.

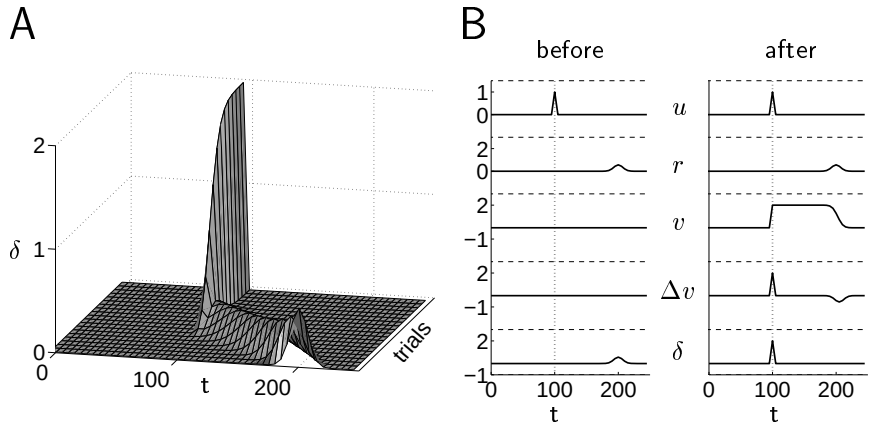

Figure 9.2 Learning to predict a reward. (A) The surface plot shows the prediction error $\delta(t)$ as a function of time within a trial, across trials. In the early trials, the peak error occurs at the time of the reward ($t = 200$), while in later trials it occurs at the time of the stimulus ($t = 100$). (B) The rows show the stimulus $u(t)$, the reward $r(t)$, the prediction $v(t)$, the temporal difference between predictions $\Delta v(t-1) = v(t) - v(t-1)$, and the full temporal difference error $\delta(t-1) = r(t-1) + \Delta v(t-1)$. The reward is presented over a short interval, and the prediction v sums the total reward. The left column shows the behavior before training, and the right column, after training. $\Delta v(t-1)$ and $\delta(t-1)$ are plotted instead of $\Delta v(t)$ and $\delta(t)$ because the latter quantities cannot be computed until time $t+1$, when $v(t+1)$ is available.

As the peak in δ moves backward from the time of the reward to the time of the stimulus, weights $w(\tau)$ for $\tau = 100, 99, \ldots$ successively grow. This gradually extends the prediction of future reward, $v(t)$, from an initial transient at the time of the reward to a broad plateau extending from the time of the stimulus to the time of the reward. Eventually, v predicts the correct total future reward from the time of the stimulus onward, and predicts the time of the reward delivery by dropping to 0 at the time when the reward is delivered. The exact shape of the ridge of activity that moves from $t = 200$ to $t = 100$ over the course of trials in figure 9.2A is sensitive to a number of factors, including the learning rate, and the form of the linear filter of equation 9.6.

Unlike the delta rule, the temporal difference rule provides an account of secondary conditioning. Suppose an association between stimulus s_1 and a future reward has been established, as in figure 9.2. When, as indicated in table 9.1, a second stimulus, s_2, is introduced before the first stimulus, the positive spike in $\delta(t)$ at the time that s_1 is presented drives an increase in the value of the weight associated with s_2, and thus establishes a positive association between the second stimulus and the reward. This exactly mirrors the primary learning process for s_1 described above. Of course, because the reward is not presented in these trials, there is a negative spike in $\delta(t)$ at the time of the reward itself, and ultimately the association between both s_1 and s_2 and the reward extinguishes.

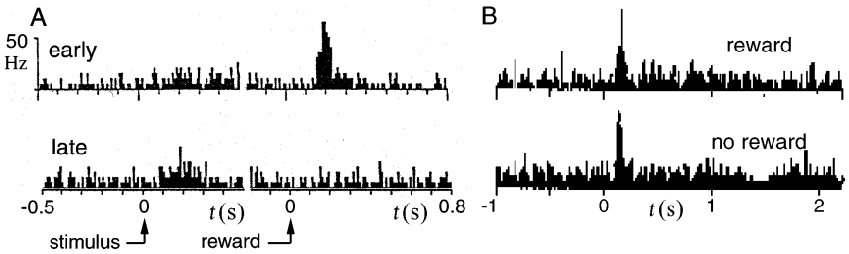

Figure 9.3 Activity of dopaminergic neurons in the VTA for a monkey performing reaction time tasks. (A) Activity of a dopamine cell accumulated over 20 trials showing the spikes time-locked to a stimulus (left panels) or to the reward (right panels) at the times marked 0. The top row is for early trials before the behavior is fully established. The bottom row is for late trials, when the monkey expects the reward on the basis of the stimulus. (B) Activity of a dopamine neuron with and without an expected reward delivery in a similar task. The top row shows the normal behavior of the cell when the reward is delivered. The bottom row shows the result of not delivering an expected reward. The basal firing rate of dopamine cells is rather low, but the inhibition at the time the reward would have been given is evident. (A adapted from Mirenowicz & Schultz, 1994; B adapted from Schultz, 1998.)

Dopamine and Predictions of Reward

The prediction error δ plays an essential role in both the Rescorla-Wagner and temporal difference learning rules, and we might hope to find a neural signal that represents this quantity. One suggestion is that the activity of dopaminergic neurons in the ventral tegmental area (VTA) in the midbrain plays this role.

ventral tegmental area VTA

There is substantial evidence that dopamine is involved in reward learning. Drugs of addiction, such as cocaine and amphetamines, act partly by increasing the longevity of the dopamine that is released onto target structures such as the nucleus accumbens. Other drugs, such as morphine and heroin, also affect the dopamine system. Further, dopamine delivery is important in self-stimulation experiments. Rats will compulsively press levers that cause current to be delivered through electrodes into various areas of their brains. One of the most effective self-stimulation sites is the medial forebrain ascending bundle, which is an axonal pathway. Stimulating this pathway is likely to cause increased delivery of dopamine to the nucleus accumbens and other areas of the brain, because the bundle contains many fibers from dopaminergic cells in the VTA projecting to the nucleus accumbens.

dopamine

In a series of studies by Schultz and his colleagues (Schultz, 1998), monkeys were trained through instrumental conditioning to respond to stimuli such as lights and sounds in order to obtain food and drink rewards. The activities of cells in the VTA were recorded while the monkeys learned these tasks. Figure 9.3A shows the activity of a dopamine cell at two times during learning. The figure is based on a reaction time task in which the monkey keeps a finger resting on a key until a sound is presented. The

monkey then has to release the resting key and press another one to get a fruit juice reward. The reward is delivered a short time after the second key is pressed. The upper plot of figure 9.3A shows the response of a cell in early trials. The cell responds vigorously to the reward, but only fires a little above baseline in response to the sound. The lower plot shows the response after a moderate amount of training. Now the cell responds to the sound, but not to the reward. The responses show a distinct similarity to the plots of $\delta(t)$ in figure 9.2.

The similarity between the responses of the dopaminergic neurons and the quantity $\delta(t)$ suggests that their activity provides a prediction error for reward, i.e. an ongoing difference between the amount of reward that is delivered and the amount that is expected. Figure 9.3B provides further evidence for this interpretation. It shows the activity of a dopamine cell in a task similar to that of figure 9.3A. The top row of this figure shows normal performance, and is just like the bottom row of figure 9.3A. The bottom row shows what happens when the monkey is expecting reward but it is not delivered. In this case, the cell's activity is inhibited below baseline at just the time it would have been activated by the reward in the original trials. This is in agreement with the prediction error interpretation of this activity.

Something similar to the temporal difference learning rule could be realized in a neural system if the dopamine signal representing δ acts to gate and regulate the plasticity associated with learning. We discuss this possibility further in a later section.

9.3 Static Action Choice

policy

In classical conditioning experiments, rewards are directly associated with stimuli. In more natural settings, rewards and punishments are associated with the actions an animal takes. Animals develop policies, or plans of action, that increase reward. In studying how this might be done, we consider two different cases. In static action choice, the reward or punishment immediately follows the action taken. In sequential action choice, rewards may be delayed until several actions are completed.

foraging

As an example of static action choice, we consider bees foraging among flowers in search of nectar. We model an experiment in which single bees forage under controlled conditions among blue and yellow artificial flowers (small dishes of sugar water sitting on colored cards). In actual experiments, the bees learn within a single session (involving visits to 40 artificial flowers) about the reward characteristics of the blue and yellow flowers. All else being equal, they preferentially land on the color of flower that delivers more reward. This preference is maintained over multiple sessions. However, if the reward characteristics of the flowers are interchanged, the bees quickly swap their preferences.